

SIMATS ENGINEERING

ASSESSMENT TOOL 3: Comparative Analysis

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA05

COURSE OUTCOME COVERED: CO4: Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics. (BTL 5)

Assessment Tool Weightage: 10%

Compute the required performance measures from the given data and compare the techniques based on the results. Evaluate and justify your conclusion using the calculated values.

1. Detection Performance Comparison

Two object detection models produce the following results:

Model	True Positives	False Positives	False Negatives
A (Viola-Jones)	80	20	30
B (YOLO)	90	15	20

- (a) Compute precision and recall for both models.
- (b) Compare their performance and state which model is better for object detection.

Working:

$$\text{Precision} = TP / (TP + FP) \quad \text{Recall} = TP / (TP + FN)$$

Step 1: Model A → Precision = $80 / (80 + 20) = 80 / 100 = 0.80$ (80%)

Step 2: Model A → Recall = $80 / (80 + 30) = 80 / 110 = 0.727$ (72.7%)

Step 3: Model B → Precision = $90 / (90 + 15) = 90 / 105 = 0.857$ (85.7%)

Step 4: Model B → Recall = $90 / (90 + 20) = 90 / 110 = 0.818$ (81.8%)

Computed Results:

Model	Precision	Recall
A (Viola-Jones)	80.0%	72.7%
B (YOLO)	85.7%	81.8%

Reading the Numbers:

Model B outscores Model A on both measures — its precision is roughly 6 points higher, meaning fewer of its positive detections are false alarms, and its recall is about 9 points higher, meaning it misses fewer actual objects. Model A's lower recall (72.7%) shows that nearly 3 in 10 real objects go undetected, a meaningful gap for a detection task.

Final Verdict: Model B (YOLO) is the better detector — it leads on both precision and recall, so there is no trade-off to weigh here.

2. Motion Estimation Error Analysis

Two motion estimation techniques produce displacement errors (in pixels):

Frame	Method A	Method B
1	2	3
2	3	5
3	2	4

- (a) Compute average error for both methods.
- (b) Compare and evaluate which method is more accurate.

Working:

Average Error = (Sum of frame errors) / (Number of frames)

Step 1: Method A $\rightarrow (2 + 3 + 2) / 3 = 7 / 3 = 2.33$ px

Step 2: Method B $\rightarrow (3 + 5 + 4) / 3 = 12 / 3 = 4.00$ px

Computed Results:

Method	Average Displacement Error (px)
Method A	2.33
Method B	4.00

Reading the Numbers:

Method A's average error of 2.33 px is noticeably lower than Method B's 4.00 px across every one of the three frames, not just on average — Method A stays ahead frame-by-frame (2 vs 3, 3 vs 5, 2 vs 4). A lower displacement error means the estimated motion vectors sit closer to the true pixel displacement, which translates directly into cleaner tracking.

Final Verdict: Method A is the more accurate motion estimator, with roughly 42% less average error than Method B.

3. Optical Flow Magnitude Comparison

Optical flow magnitudes (in pixels/frame):

Region	Method A	Method B
R1	5	8
R2	6	10

Region	Method A	Method B
R3	5	9

- (a) Compute average motion magnitude for both methods.
- (b) Compare and evaluate which method captures stronger motion.

Working:

Average Magnitude = (Sum of region magnitudes) / (Number of regions)

Step 1: Method A $\rightarrow (5 + 6 + 5) / 3 = 16 / 3 = 5.33$ px/frame

Step 2: Method B $\rightarrow (8 + 10 + 9) / 3 = 27 / 3 = 9.00$ px/frame

Computed Results:

Method	Average Motion Magnitude (px/frame)
Method A	5.33
Method B	9.00

Reading the Numbers:

Method B records a higher magnitude in every region (8 vs 5, 10 vs 6, 9 vs 5), pushing its average to 9.00 px/frame against Method A's 5.33 px/frame — roughly 69% stronger. This means Method B is more sensitive to motion and registers larger displacement per frame. Whether that is desirable depends on the application: it is useful for capturing fast or subtle motion, but a consistently higher reading can also signal that Method B is more prone to amplifying noise, so it should not automatically be read as "more correct."

Final Verdict: Method B captures stronger motion in absolute magnitude, making it the better choice when sensitivity to motion is the priority, while Method A remains the more conservative, potentially more stable estimator.

4. Depth Estimation Comparison (Stereo vs SfM)

Estimated depth values (in meters); actual depths are A = 2, B = 3, C = 4:

Object	Stereo Vision	Structure from Motion
A	2.0	2.5
B	3.0	3.8
C	4.0	5.0

- (a) Compute absolute error for both methods.
- (b) Compare and evaluate which method is more accurate.

Working:

Absolute Error = | Estimated Value – Actual Value |

Step 1: Stereo Vision $\rightarrow |2.0-2| = 0, |3.0-3| = 0, |4.0-4| = 0$

Step 2: Structure from Motion $\rightarrow |2.5-2| = 0.5, |3.8-3| = 0.8, |5.0-4| = 1.0$

Step 3: Average absolute error $\rightarrow \text{Stereo} = 0 / 3 = 0.0 \text{ m}; \text{SfM} = (0.5+0.8+1.0) / 3 = 2.3 / 3 = 0.77 \text{ m}$

Computed Results:

Object	Stereo Error (m)	SfM Error (m)
A	0.0	0.5
B	0.0	0.8
C	0.0	1.0
Average	0.0	0.77

Reading the Numbers:

Stereo Vision matches the actual depth exactly for all three objects, giving a zero error across the board. Structure from Motion consistently overestimates depth, and the gap widens as distance increases — the error grows from 0.5 m at 2 m to a full 1.0 m at 4 m, suggesting its error scales with depth rather than staying constant.

Final Verdict: Stereo Vision is clearly the more accurate method here, with zero measured error against SfM's 0.77 m average error — though this comes at the cost of needing a second camera, which is a real deployment trade-off, not just a reasoning point.

5. Pattern Recognition Accuracy Comparison

Two recognition systems produce results:

System	Correct Predictions	Total Samples
Traditional	85	100
Deep Learning	92	100

- (a) Compute accuracy for both systems.
- (b) Compare and evaluate which system performs better.

Working:

$$\text{Accuracy} = \text{Correct Predictions} / \text{Total Samples} \times 100$$

Step 1: Traditional $\rightarrow 85 / 100 \times 100 = 85\%$

Step 2: Deep Learning $\rightarrow 92 / 100 \times 100 = 92\%$

Computed Results:

System	Accuracy
Traditional	85%

System	Accuracy
Deep Learning	92%

Reading the Numbers:

Deep Learning correctly classifies 7 more samples out of every 100 than the Traditional system — a 7-percentage-point gap that, at scale, represents a substantial reduction in misclassifications. Since both systems were evaluated on the same 100 samples, the comparison is direct and not skewed by differing sample sizes.

Final Verdict: Deep Learning performs better on this dataset, with 92% accuracy against the Traditional system's 85%.

Code of Conduct:

I G.Joel Richard (192524054) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand